Random Variables

math-foundations / node 22

First, an example

Before the abstract definition, picture a single fair die roll. Take the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$, with $\mathcal{F}$ the power set and P the uniform measure. Define two functions on Ω:

$$X(\omega) = \omega, \qquad Y(\omega) = \begin{cases} 1 & \text{if } \omega \text{ is even,} \\ 0 & \text{otherwise.} \end{cases}$$

Read aloud: "X of omega equals omega; Y of omega equals one if omega is even, zero otherwise." Here X takes values in $\{1, \ldots, 6\}$ (the number shown), while Y takes values in $\{0, 1\}$ (an *indicator* of the event "even"). The event "$X = 4$" is the singleton $\{4\} \subset \Omega$, which lies in $\mathcal{F}$ — so X qualifies as a random variable. Keep this picture in mind as we generalize below.

1 Setting

Fix a probability space $(\Omega, \mathcal{F}, P)$ and write $\mathcal{B}(\mathbb{R})$ for the Borel σ-algebra of $\mathbb{R}$ (the smallest σ-algebra containing the open intervals).

Definition 1 (Random variable)**.** *A function $X : \Omega \to \mathbb{R}$ is a* random variable *if*

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \ \in \ \mathcal{F} \qquad \textit{for every } B \in \mathcal{B}(\mathbb{R}).$$

Read aloud: *"the pre-image of B under X — that is, the set of omega in Omega with X of omega in B — is an element of script F, for every Borel set B in R."*

Remark 2. *Equivalently, X is a random variable iff $\{X \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. This follows because $\{(-\infty, x] : x \in \mathbb{R}\}$ generates $\mathcal{B}(\mathbb{R})$ and X^{-1} commutes with countable unions, intersections, and complements.*

Definition 3 (Distribution function)**.** *The* cumulative distribution function *(CDF) of a random variable X is*

$$F_X : \mathbb{R} \to [0, 1], \qquad F_X(x) := P(X \leq x) = P\big(X^{-1}((-\infty, x])\big).$$

Read aloud: *"F sub X maps R to the unit interval; F sub X of x is defined as the probability that X is at most x, equivalently the P-measure of the pre-image of negative-infinity to x under X." The function F_X is non-decreasing, right-continuous, with $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$.*

2 Algebra of random variables

We now prove that the sum of two random variables is again a random variable. This is the *prototype* for showing that products, maxima, minima, and limits of random variables remain random variables.

Lemma 4 (Pre-image of an open set)**.** *Let $X : \Omega \to \mathbb{R}$ satisfy $\{X \le x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$. Then for every open set $U \subseteq \mathbb{R}$, $X^{-1}(U) \in \mathcal{F}$.*

Proof. Every open subset of $\mathbb{R}$ is a countable union of open intervals. For an interval (a, b),

$$X^{-1}((a,b)) = \{X < b\} \cap \{X > a\} = \Big(\bigcup_{n=1}^{\infty} \{X \le b - 1/n\}\Big) \cap \{X \le a\}^c,$$

Read aloud: "the pre-image of the open interval (a, b) under X equals the intersection of X-less-than-b and X-greater-than-a, which equals the countable union over n of X-at-most-b-minus-one-over-n, intersected with the complement of X-at-most-a." which is a countable Boolean combination of sets in $\mathcal{F}$, hence in $\mathcal{F}$. Taking countable unions preserves this. □

Theorem 5 (Sum of random variables is a random variable)**.** *If $X, Y : \Omega \to \mathbb{R}$ are random variables, then $Z := X + Y$ is a random variable.*

Proof. By the remark, it suffices to show $\{Z \le c\} \in \mathcal{F}$ for every $c \in \mathbb{R}$. Equivalently, since $\{Z \le c\} = \{Z < c\}^c \cup \{Z = c\}$ and complements stay in $\mathcal{F}$, we instead exhibit $\{Z < c\}$ as a measurable set — this is the cleaner form and uses Lemma 4 indirectly.

Observe that $X(\omega) + Y(\omega) < c$ holds iff there exists a rational q with

$$X(\omega) < q \quad \text{and} \quad Y(\omega) < c - q.$$

Read aloud: "X of omega is less than q, and Y of omega is less than c minus q." ($\Rightarrow$): If $X(\omega) + Y(\omega) < c$, set $\varepsilon = c - X(\omega) - Y(\omega) > 0$; by density of $\mathbb{Q}$ in $\mathbb{R}$, pick $q \in \mathbb{Q}$ with $X(\omega) < q < X(\omega) + \varepsilon/2$. Then $q < X(\omega) + \varepsilon/2$ gives $c - q > c - X(\omega) - \varepsilon/2 = Y(\omega) + \varepsilon/2 > Y(\omega)$. ($\Leftarrow$): If $X(\omega) < q$ and $Y(\omega) < c - q$, add the inequalities.

Hence

$$\{Z < c\} \;=\; \bigcup_{q \in \mathbb{Q}} \Big(\{X < q\} \cap \{Y < c - q\}\Big).$$

Read aloud: "the event Z-less-than-c equals the union over rational q of the intersection of X-less-than-q and Y-less-than-c-minus-q." The collection $\mathbb{Q}$ is countable. Each $\{X < q\} = X^{-1}((-\infty, q))$ and each $\{Y < c - q\} = Y^{-1}((-\infty, c - q))$ is the pre-image of an open set, hence in $\mathcal{F}$ by Lemma 4. A countable union of intersections of $\mathcal{F}$-sets is in $\mathcal{F}$, so $\{Z < c\} \in \mathcal{F}$. Finally $\{Z \le c\} = \bigcap_{n=1}^{\infty} \{Z < c + 1/n\} \in \mathcal{F}$, proving the theorem. □

Remark 6. *The same density-of-rationals trick proves $XY, \max(X, Y), \min(X, Y)$ are random variables. Iterating gives that every polynomial in finitely many random variables is a random variable, and combining with Theorem 3 of the README extends the result to limits.*